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OPEN

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^{18}F-DPA-714 PET/CT Macrophage Imaging in Triple-Negative Breast Cancers (EITHICS)

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Abstract: Tumor-associated macrophages are targets of interest in triple-negative breast cancer (TNBC). The translocator protein 18 kDa (TSPO) is a sensitive marker for macrophages and holds potential relevance in TNBC stratification. This pilot prospective study (EITHICS, NCT04320030) aimed to assess the potential of TSPO PET/CT imaging using ^{18}F-DPA-714 in primary TNBC, compared with immunohistochemistry, autoradiography, and TSPO polymorphism.

Patients and Methods: Thirteen TNBC patients were included. They underwent TSPO genotyping (HAB, MAB, LAB), ^{18}F-FDG PET/CT, and breast MRI. Semiquantitative PET parameters were computed. VOIs were defined on the tumor lesion, healthy breast tissue, and pectoral muscle to obtain SUV, tumor-to-background ratio (TBR), and time-activity curves (TACs). Additionally, immunohistochemistry, ^{3}H-DPA-714, and ^{3}H-PK-11195 autoradiography were conducted.

Results: The majority of TNBC tumors (11/13, 84%) had a preponderance of M2-polarized macrophages with a median proportion of 82% (range, 44%–94%). ^{18}F-DPA-714 PET/CT clearly identified TNBC tumors with an excellent TBR. Three distinct patterns of ^{18}F-DPA-714 TACs were identified, categorized as "above muscular," "equal to muscular," and "below muscular" with reference to the muscular background. For the "above muscular" group (2 HAB and 2 MAB), "equal muscular" group (3 HAB, 3 MAB, and 1 LAB), and "below muscular" group (1 LAB and 1 MAB), tumor TACs showed a ^{18}F-DPA-714 accumulation slope of 1.35, 0.62, and 0.22, respectively, and a median SUV_{mean} of 4.02 (2.09–5.31), 1.66 (0.93–3.07), and 0.61 (0.43–1.02).

Conclusions: This study successfully demonstrated TNBC tumor targeting by ^{18}F-DPA-714 with an excellent TBR, allowing to stratify 3 patterns of uptake potentially influenced by the TSPO polymorphism status. Further studies in larger populations should be performed to evaluate the prognostic value of this new biomarker.

Key Words: triple-negative breast cancer, PET/CT, TSPO, ^{18}F-DPA-714, inflammation, tumor-associated macrophage

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Breast cancer (BC) is a heterogeneous disease with several subtypes including triple-negative breast cancer (TNBC), which has a poor prognosis.[1] This classification is defined by the absence of estrogen and progesterone receptor expression, and the nonoverexpression or amplification of human epidermal growth factor receptor 2 (HER2). Recently, the combination of immune checkpoint inhibitors targeting programmed death 1 (PD-1) or programmed death ligand 1 (PD-L1) with chemotherapy has improved pathological complete response rates (neoadjuvant treatment) and long-term outcomes in nonmetastatic (neoadjuvant and adjuvant treatment) and metastatic TNBC.[2–5]

Despite these advances, a subset of TNBC patients does not exhibit a favorable response to these therapeutic regimens. Consequently, there is a compelling need for strategies that can predict TNBC status to improve the outcome of immune-targeted therapies.

Immunohistochemical (IHC) approaches have shown that the TNBC tumor microenvironment (TME) is characterized by high expression of growth and migration-promoting molecules, and by a greater number of immune cells, especially by tumor-associated macrophages (TAMs).[6] In fact, several studies have shown that the TME plays a key role in tumor development and response to treatment,[7] and TAMs may play a key role in tumor progression and treatment-resistant mechanisms.[8] Indeed, TAMs have been shown to modulate PD-1/PD-L1 expression and increase PD-L1 expression via cytokine secretion.[9] It is therefore essential to study TME in TNBC and its interaction with tumor cells, but currently, we only have IHC approaches available, which require biopsy analysis, and it would be beneficial to have a comparable noninvasive characterization method.

Molecular imaging could play a role in the noninvasive study of TNBC TME. Indeed, the translocator protein 18 kDa (TSPO) is expressed by TAMs and can be targeted by PET using the radiopharmaceutical ^{18}F-DPA-714 (N,N-diethyl-2-[4-(2-fluoroethoxy) phenyl]-5,7-dimethylpyrazolo[1,5-a]pyrimidine-3-acetamide labeled with fluorine 18). TSPO, originally described as the peripheral benzodiazepine receptor, is an intracellular protein involved in many biological functions including apoptosis, cell proliferation, and immunomodulation.[10] TSPO is predominantly localized on the outer mitochondrial membrane, but it is also found around the nuclei of certain tumor cells, particularly in BC.[11] Although this ubiquitous protein plays essential physiological roles, it is also involved in various pathological phenomena, including tumor progression. Increased TSPO protein expression has been observed in numerous human tumor cells and tissues, particularly in hormone receptor-negative BCs, where TSPO expression appears to correlate with tumor aggressiveness and progression.[12–14] Additionally, TSPO serves as an in vivo biomarker of peripheral inflammation and is particularly expressed in myeloid lineage cells, such as macrophages.[15,16] TSPO PET/CT has previously been used as a biomarker of brain neuroinflammation,[17] for assessing joint inflammation in rheumatoid arthritis[18] or for evaluating lung inflammation.[19]

Overall, TSPO is emerging as a promising in vivo prognostic and therapeutic target. A preclinical study using human breast tumor cell transplantation models in mice and ^{18}F-DPA-714-based imaging has validated TSPO as a biomarker of tumor cells and TME, particularly for TAMs.[20]

The aim of this pilot feasibility study (EITHICS) was to compare ^{18}F-DPA-714 PET/CT imaging with tumor data obtained by IHC, autoradiography, and TSPO polymorphism to investigate the primary operable TNBC tumor and its TME.

PATIENTS AND METHODS

Design and Population

This study (EITHICS) is a prospective, multicenter, nonrandomized phase II trial. Patients were enrolled between June 2020 and May 2021 in 2 hospitals in France. Ethical approval was obtained from the Ile de France II Review Board (19.10.17.46746). All patients signed a written informed consent prior to participation in this study (2019-10-00030).

All patients had TNBC (estrogen and progesterone receptors both <10%, and HER2 not amplified or overexpressed) with a surgical indication. Patients with inflammatory or metastatic TNBC were excluded. Patients were also excluded if they had received any anticancer treatment (chemotherapy, immunotherapy, hormone therapy, and external beam radiotherapy), antibiotics, and/or anti-inflammatory drugs (steroidal and/or nonsteroidal) in the 30 days prior to inclusion.

The study design is shown in Supplemental Figure 1, http://links.lww.com/CNM/A476. Initially, a blood sample was taken from each patient for TSPO gene polymorphism detection (rs6971). Because of the time required to obtain the genotyping results, a pooled and staggered analysis was chosen at the end of the study. The TSPO polymorphism (rs6971) predicts the binding affinity to second-generation TSPO tracers,[21] including ^{18}F-DPA-714, and patients included were potentially high, medium, or low affinity binders (HAB, MAB, and LAB). The TSPO gene polymorphism status was determined for all patients.

Whole-body ^{18}F-FDG PET/CT and multiparametric MRI with bilateral breast and axillary diffusion sequence were performed initially. For the ^{18}F-DPA-714 PET/CT scan, 3 sequential acquisitions were performed. After surgery, tumor tissue was examined by IHC analysis, and in vitro autoradiography using ^{3}H-DPA-714 and ^{3}H-PK-11196 ((R)-*N*-methyl-*N*-(1-methylpropyl)-1-(2-chlorophenyl)isoquinoline-3-carboxamide).

Radiosynthesis of ^{18}F-DPA-714

DPA-714 was labeled with fluorine-18 at its 2-fluoroethyl moiety, following nucleophilic substitution of the corresponding tosylate analog, according to slight modifications of previously reported procedures.[22] The formulation of ^{18}F-DPA-714 provided a sterile injectable solution of isotonic sodium chloride with ethanol at a mass percentage of <8% for total injected volumes (ranging from 3 to 5 mL), in accordance with the European Pharmacopeia. The mean specific activity of ^{18}F-DPA-714 obtained was 251.65 ± 60.23 GBq/μmol.

Imaging Acquisition Parameters

^{18}F-DPA-714 Images

The radiopharmaceutical activity delivered was 3.5 MBq/kg ± 10%, in agreement with the literature.[17] PET images were acquired with 2 types of cameras: Biograph Vision 450 and Biograph mCT (Siemens Healthcare, Erlangen, Germany). A first series of dynamic PET images in the procubitus position was acquired in list mode for 30 minutes (30 images of 1 minute each), centered on the thorax and axillary area, starting at the time of injection. The PET acquisition was combined with a low-dose CT scan centered on the chest. Then, a 15-minute thoracic static acquisition in procubitus was performed at 45 minutes postinjection. Finally, a decubitus whole-body acquisition from the head to the mid-thighs was performed 60 minutes postinjection, 120 kV, 60 mAs, with 5-mm slice thickness (16 × 1.2 mm), and a pitch of 0.85.

MRIs

Procubitus breast MRI was performed with a dedicated 16-channel breast coil on a 1.5 T MAGNETOM SOLA (Siemens Healthcare GmbH, Erlangen, Germany). The protocol included preinjection axial T1 TSE and T2 TSE, 3 mm. After automatic injection of Clariscan GE 0.5 mmol/mL (gadoteric acid), 0.2 mL/kg, followed by a 20-mL saline bolus, a DIXON VIBE 3D iso 0.9 dynamic axial acquisition was performed, followed by 4 postinjection series.

Data Analysis

For each patient, on ^{18}F-DPA-714 PET/CT fusion images (procubitus dynamic images sum), a volume of interest (VOI) on the breast tumor lesion was defined using a thresholding method relative to 40% tumor SUV_{max}, using 3D Slicer software (version 5.2), then transposed onto the healthy contralateral mammary gland and contralateral pectoral muscle to obtain time-activity curves (TACs). These TACs allowed the analysis of ^{18}F-DPA-714 kinetics over the first 30 minutes of PET acquisition for the different tissues. In addition, a VOI was manually traced around the tumor, including the peritumoral tissue. The ratios (tumor-to-background ratio or TBR) between the mean activity SUV_{max} of the breast tumor lesion VOI and the healthy breast tissue VOI were calculated. The SUV_{max} values of the tumor and contralateral breast tissue were determined manually by 2 experienced nuclear medicine physicians using the syngo.via 3D postprocessing software (Siemens Healthcare GmbH, Erlangen, Germany).

Histologic and IHC Evaluations

Thirteen cases of archival formalin-fixed and paraffin-embedded tissues of the internal cohort were evaluated on full sections for histological evaluation. All tumor tissue samples were surgically collected and were fixed in 10% neutral buffered formalin for standard histological analysis and IHC. IHC was performed using the Leica BOND-MAX platform. Details of the antigen retrieval technique and dilution of primary antibodies CD68 and CD163 are listed in Supplemental Table 1, http://links.lww.com/CNM/A477. The number of CD68-positive and CD163-positive cells per mm^2 was evaluated in hot spots. Tumor-infiltrating lymphocytes and CD138-positive plasma cells were assessed according to recommendations of an international working group.[23,24]

Comparative Evaluation of TSPO Radioligands ^{3}H-DPA-714 and ^{3}H-PK-11195 by In Vitro Autoradiographic Binding on TN Tumor Slices

The density-specific TSPO binding sites were measured by in vitro autoradiographic assessments using ^{3}H-DPA-714 and ^{3}H-PK-11195 on adjacent TNBC tumor sections from 10 of the 13 included patients (the surgical specimen obtained from tumor resection surgery was insufficient for the 3 other patients). ^{3}H-DPA-714 was prepared according to Damont et al,[25] (molar activity 2.1 GBq/μmol, CEA, Orsay, France), and ^{3}H-PK-11195 was commercially purchased (molar activity 3.06 GBq/μmol; Perkin Elmer, Norwalk, CT). Frozen samples were sliced into 16-μm-thick sections using a cryostat at −20°C (CM 3050S; Leica, Germany). Sections on gelatinized slides were stored at −80°C until required. Tumor sections were allowed to equilibrate at room temperature (RT) for 3 hours, then incubated with 1 nM of labeled ligand (^{3}H-DPA-714 or ^{3}H-PK-11195) in 50 mmol/L Tris–HCl buffer pH 7.4 at RT for 60 minutes. Nonspecific binding was assessed by incubation of adjacent sections in the presence of 1 μM of cold PK-11195 (Sigma Aldrich, Lyon, France). Sections were rinsed twice in ice cold buffer (4°C) for 5 minutes, then briefly in distilled water at 4°C and dried at RT. Slides were made conductive by applying a metal electric tape (3M; Euromedex, Souffelweyersheim, France) and then were placed in the gas chamber of the b-Imager 2000 (Biospace Lab, Paris, France). Acquisitions were compiled over a period of 3 hours. Hyperintense binding areas were manually contoured on each total binding section, and each defined region of interest was reported on adjacent slices pretreated with cold PK11195 for nonspecific binding measurement. The level of bound radioactivity was quantitated by counting the number of β-particles emitted from the delineated area (β-Vision Software; Biospace Lab, Paris, France) in the regions of interest. The radioligand signals, expressed in counts per minute per square millimeter (cpm/mm^2), were measured for 6 sections per patient using an image analyzer (M3 Vision; Biospace Instruments, Paris, France). Specific binding (SB) was determined by subtracting nonspecific binding from total binding. The ^{3}H-DPA-714 SB/^{3}H-PK-11195 SB ratio was determined for each patient to evaluate the signal intensity for each radioligand, relative to TSPO genotype.

Statistical Analysis

Qualitative factors were described by the frequency of their respective modalities and continuous factors by their mean ± standard deviation (or median, range). Post hoc subgroups were compared using Pearson χ^2 test (or Fisher test if necessary) for categorical variables and Wilcoxon test (or Kruskal-Wallis or Mann-Whitney *U* tests if necessary) for continuous variables. Correlations between continuous variables were calculated using Spearman correlation coefficient. All tests were 2-tailed, with significance set at 5%. The software used was Stata18.0 (StataCorp, College Station, TX).

RESULTS

A total of 13 TNBC patients were included in the study. Breast MRI confirmed unifocal breast involvement in all patients. The median tumor size was 18 mm (range, 9–33) with a median Ki67 of 60% (range, 13%–90%). Patient characteristics are summarized in Table 1. Eleven patients underwent lumpectomy, and 2 underwent mastectomy. Ten of 13 of patients had tumors classified as pT1, and 3/13 were classified as pT2. Lymph node involvement was histologically identified in only 1 patient.

Histological data showed that almost all patients had nonspecific infiltrating carcinoma (12/13, 92%). No adverse events occurred immediately or remotely after ^{18}F-DPA-714 injection. No relapses or deaths were observed during a median follow-up of 33.4 months (range, 27.6–38.5).

^{18}F-DPA-714 PET/CT showed significant tumor uptake with a good TBR, except in 2 patients. Median lesion SUV_{max} values were 4.73 (range, 0.98–11.09) for ^{18}F-DPA-714 PET/CT and 7.17 (range, 2.27–33.84) for ^{18}F-FDG PET/CT. All lesion SUV_{max} and SUV_{mean} values for ^{18}F-DPA-714 and ^{18}F-FDG are detailed for each patient in Supplemental Table 2, http://links.lww.com/CNM/A478. The imaging panel performed for each patient, MRI, ^{18}F-DPA-714 PET/CT, and ^{18}F-FDG PET/CT, is shown in Figure 1. In all patients, ^{18}F-DPA-714 PET/CT showed physiological uptake of bone marrow limited to the spine, spleen, myocardium, salivary glands, gallbladder, liver, and, to a lesser extent, digestive structures and muscles. Apart from these ^{18}F-DPA-714 foci and the tumor involvement, there was no evidence of other uptake, in particular in the brain or in the joints.

TABLE 1. Patient Characteristics

#	Age	Histological Type	cTNM	Surgery Type	SALND	Lesion Size (mm)	pTpN	SBR Grade	UICC Status	% ER Status	% PR Status	HER2 Status	Vascular Emboli	Ki67 (%)
1	51	NSIC	T1N0M0	Mastectomy	Yes	16	T1N0	III	I	0	0	1+	No	75
2	68	NSIC	T1N0M0	Lumpectomy	Yes	20	T1N1	II	IIA	0	0	0	No	22
3	69	NSIC	T1N0M0	Lumpectomy	Yes	18	T1N0	III	I	0	0	0	Yes	35
4	75	NSIC	T2N0M0	Lumpectomy	Yes	23	T2N0	II	IIA	5	5	0	Yes	13
5	41	NSIC	T2N1M0	Lumpectomy	Yes	13	T1N0	II	I	0	0	0	No	20
6	49	NSIC	T1N0M0	Lumpectomy	Yes	26	T2N0	III	IIA	0	0	0	No	80
7	68	NSIC	T0N0M0	Lumpectomy	Yes	16	T1N0	III	I	0	0	1+	Yes	60
8	54	NSIC	T1N0M0	Lumpectomy	Yes	10	T1N0	III	I	0	0	0	No	70
9	50	NSIC	T1N0M0	Lumpectomy	Yes	33	T2N0	III	IIA	0	0	2+ (Fish neg)	No	90
10	67	NSIC	T1N0M0	Lumpectomy	Yes	18	T1N0	III	I	0	0	0	No	40
11	26	NSIC	T1N0M0	Lumpectomy	Yes	18	T1N0	-	I	0	0	0	No	60
12	84	IPC	T1N0M0	Mastectomy	Yes	20	T1N0	III	I	0	0	0	Yes	40
13	53	NSIC	T1N0M0	Lumpectomy	Yes	9	T1N0	III	I	0	0	2+ (Fish neg)	No	60

cTNM, clinical stage TNM; SBR grade, Scarff-Bloom-Richardson grade; UICC, Union for International Cancer Control; SALND, sentinel axillary lymph node dissection; pTpN, pathological stage TNM; % ER, % estrogen receptor; % PR, % progesterone receptor; NSIC, nonspecific invasive carcinoma; IPC, invasive papillary carcinoma.

Figure 2 shows ^{18}F-DPA-714 TACs uptake for all subjects in the tumor, breast background (contralateral healthy breast) and muscular background (tumor contralateral pectoral). Three different patterns of TACs of ^{18}F-DPA-714 binding in TNBC were observed and classified as "above," "equal to," and "below" muscular background. Healthy breast TAC was almost flat (ie, baseline levels), in contrast to pectoral muscle TAC, which showed an ^{18}F-DPA-714 rapid uptake with a slope of 0.55 followed by an asymptotic appearance with a median SUV_{mean} of 1.18 (range, 0.18–3.88). Three different tumor TAC patterns were observed, and we chose to compare them with the muscular background. From the TACs obtained for the "above muscular" group (2 HAB and 2 MAB), the "equal muscular" group (3 HAB, 3 MAB, and 1 LAB), and the "below muscular" group (1 LAB and 1 MAB), the slope was calculated, and the values found were 1.35, 0.62, and 0.22, respectively. For the "above muscular," "equal muscular," and "below muscular" groups, the plateau of each curve had a median SUV_{mean} of 4.02 (range, 2.09–5.31), 1.66 (range, 0.93–3.07), and 0.61 (range, 0.43–1.02), respectively.

For the 13 patients considered together, the calculated TBR (SUV_{max} TNBC/SUV_{max} contralateral breast) at the 15th min postinjection was 11.71 (range, 2.99–51.15). For each "above," "equal," and "below" group, the TBRs were 19.94 (range, 17.61–51.15), 10.70 (range, 7.32–17.96), and 3.77 (range, 2.99–4.54), respectively. These data are supported by the TACs, which showed a significant difference between the breast background and the TNBC curves.

Peritumoral macrophage infiltration was assessed by IHC using the CD68 marker (Fig. 3). The percentage of M2-polarized macrophages was estimated using the CD163 marker (Table 2). The majority of TNBC tumors (11/13, 84%) showed a preponderance of M2-polarized macrophages with a median of 82% (range, 44%–94%). No significant correlation was observed between the semiquantitative TNBC lesion parameters (SUV_{max}, SUV_{mean}, and %ID/g) of ^{18}F-DPA-714 PET/CT and the number of CD68 macrophages per field estimated by IHC ($P = 0.311$) or the percentage of M2-polarized macrophages estimated by IHC using the CD163 marker ($P = 0.122$).

The results for the TSPO polymorphism screening are shown in Table 3 and include the comparative evaluation of TSPO radioligands ^{3}H-DPA-714 and ^{3}H-PK-11195 by in vitro autoradiographic binding on TNBC tumor slices. It is noteworthy that PK11195 is not sensitive to TSPO polymorphism, unlike second-generation TSPO radioligands, including DPA-714.[21] First, the homozygous rs6971 polymorphism (A/A) was found in 2 of 13 (15%) patients, which is related to a "low binding" status (LAB) for the second-generation TSPO radiopharmaceuticals as ^{18}F-DPA-714. The other patients were either heterogeneous (MAB) (A/G; n = 6; 46%) or lacked polymorphism (HAB) (G/G; n = 5; 39%). Binding of ^{3}H-PK-11195 and ^{3}H-DPA-

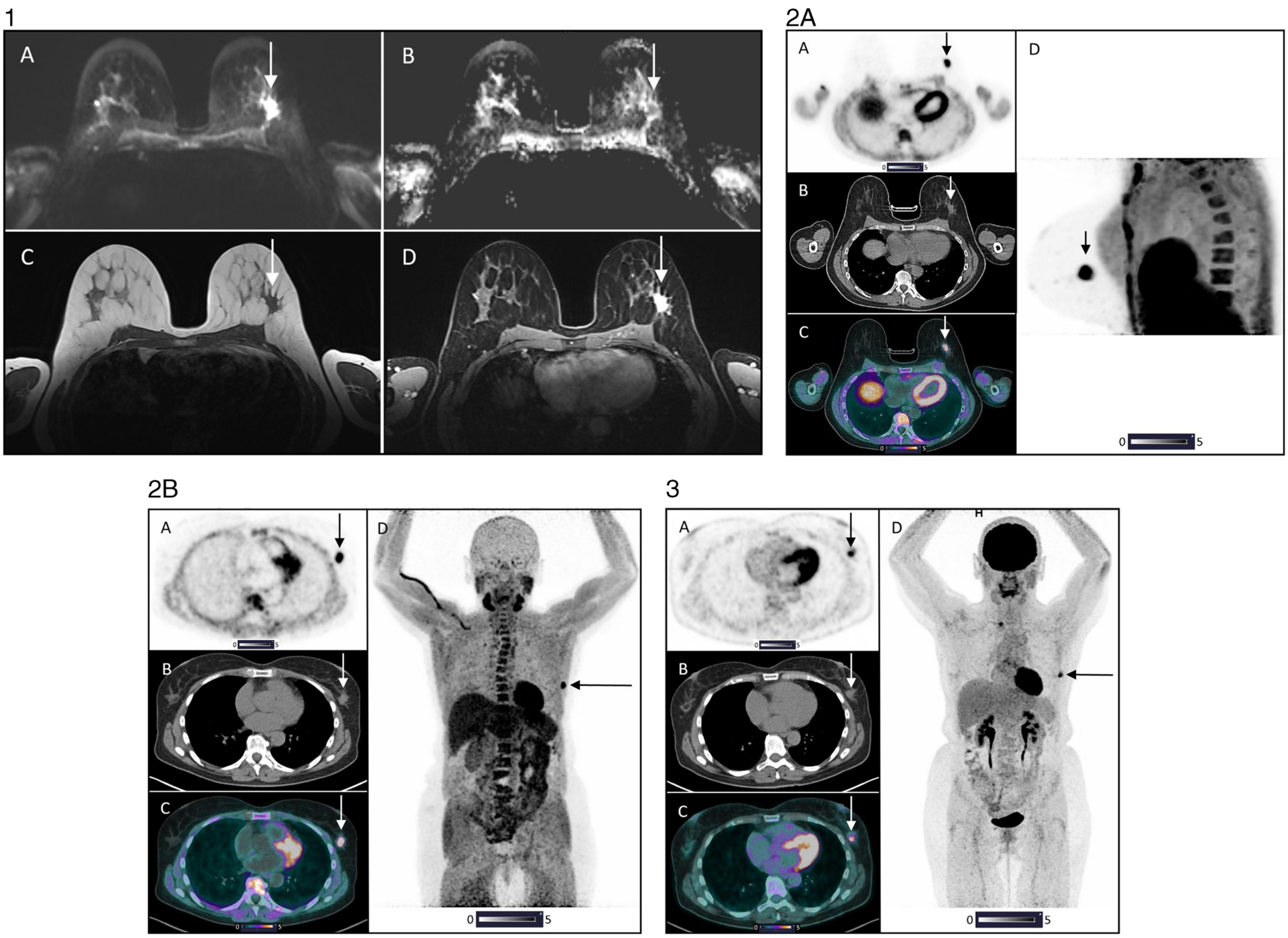

FIGURE 1. Patient 4 iconography: mpMRI (**1**) dynamic early ^{18}F-DPA-714 PET/CT acquisition on procubitus (**2A**), whole-body ^{18}F-DPA-714 PET/CT on decubitus at 60 minutes postinjection (**2B**), and whole-body ^{18}F-FDG PET/CT (**3**).

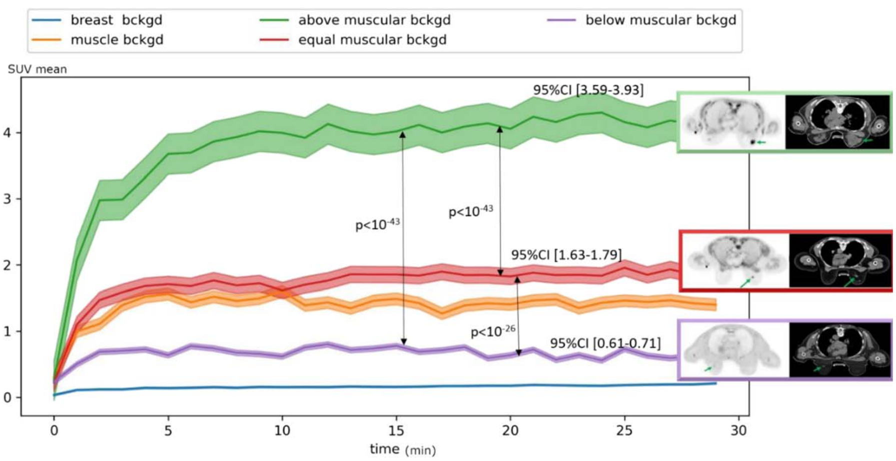

FIGURE 2. Time-activity curves of ^{18}F-DPA-714 binding on breast background (contralateral healthy mammary gland) and muscular background (pectoral). Three different patterns of TACs of ^{18}F-DPA-714 binding on TNBC were observed: "above," "equal to," and "below" muscular background.

714 to TSPO was compared using autoradiography on TNBC tumor sections from 10 patients (2 HAB, 6 MAB, and 2 LAB). Autoradiography quantitative results revealed that SB of the radioligand was significantly higher with ^{3}H-PK-11195 than with ^{3}H-DPA-714 (P < 0.01). In addition, no statistically significant correlation was found with DPA-714 between in vivo PET/CT SUV_{max} and SUV_{mean} and in vitro SB intensity. Interestingly, and despite the small sample size, the ^{3}H-DPA-714 SB/^{3}H-PK-11195 SB ratio was significantly lower when comparing LAB patients with HAB and MAB patients pooled together (P < 0.05).

Qualitative autoradiographic analysis showed that uptake in some tumor sections was more intense at the tumor periphery than in the center, particularly with ^{3}H-PK-11195, as if uptake was stronger in the TME than in the tumor cells themselves.

DISCUSSION

To our knowledge, this proof-of-concept clinical study is the first to assess the feasibility and relevance of evaluating TNBC primary tumors at the time of diagnosis using PET/CT with ^{18}F-DPA-714, a second-generation TSPO tracer.

Two important findings should be highlighted: first, ^{18}F-DPA-714 PET/CT clearly identified TNBC tumors with an excellent TBR, and the TACs generated for each patient showed that ^{18}F-DPA-714 uptake distinguished 3 significantly different patterns. Second, the results of the TSPO polymorphism and the comparative evaluation of the TSPO radioligands ^{3}H-DPA-714 and ^{3}H-PK-11195 by in vitro autoradiography binding to TNBC tumor sections showed that the ratio (^{3}H-DPA-714 SB)/(^{3}H-PK-11195 SB) was significantly lower when comparing LAB patients with HAB and MAB patients pooled together, which, despite the small sample size, seems to be consistent with the PET/CT data.

With regard to the ^{18}F-DPA-714 biodistribution, similar to Arlicot et al,[17] we observed intense physiological uptake in the myocardium, spleen, bone marrow (spine), liver, kidneys, salivary glands, adrenals, and, to a lesser extent, muscular structures. We also observed intense binding in the gallbladder and digestive structures, which is compatible with partial hepatobiliary elimination of the radiotracer, as suggested by the same team.[17] Despite high physiological uptake, there may be a gradient of uptake in favor of the neoplastic lesion, making it possible to distinguish between physiology and neoplasia. We were unable to document this possible aspect as all our patients were imaged in the early stages of their disease. However, all these physiological foci could limit the usefulness of ^{18}F-DPA-714 PET/CT for theranostic purposes.

Qualitative analysis of ^{18}F-DPA-714 PET/CT showed significant uptake in breast tumor lesions with an excellent TBR, except in 2 patients. One of these patients had a TSPO LAB polymorphism, described in the literature outside oncology, particularly in neurodegenerative diseases, as corresponding to 10% of the population. The TSPO LAB polymorphism appeared to be nonbinding to the second-generation TSPO tracer such as ^{18}F-DPA-714.[17,26] Indeed, our autoradiographic results suggested reduced binding of ^{18}F-DPA-714 to TSPO in TNBC patients with an LAB polymorphism compared with MAB + HAB patients, based on the ratio (^{3}H-DPA-714 SB)/(^{3}H-PK-11195 SB), which was significantly lower in LAB

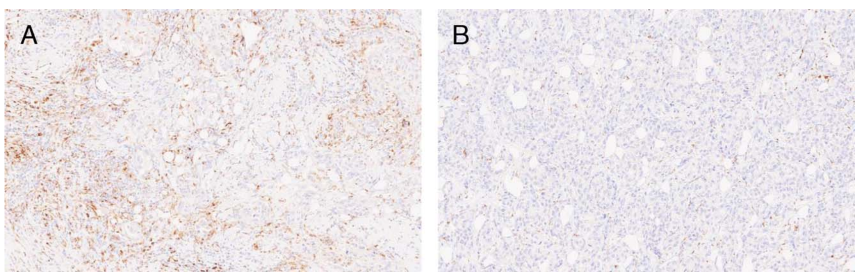

FIGURE 3. Immunohistochemical staining of 2 TNBC cases. Anti-CD68 staining of breast tumor biopsy showing high (**A**) and low (**B**) CD68 macrophage abundance. Magnification ×100.

TABLE 2. Intratumoral Immunological Quantification of CD163 and CD68-Positive Macrophages

#	Nb CD68 Positive-Cells per mm^2 (M1 and M2 Type Macrophages)	Nb CD163 Positive-Cells per mm^2 (%) (M2 Type Macrophages)
1	88	83 (94)
2	27	17 (63)
3	62	51 (82)
4	9	4 (44)
5	47	39 (83)
6	77	63 (82)
7	60	52 (87)
8	12	6 (50)
9	85	74 (87)
10	114	102 (89)
11	43	32 (74)
12	24	15 (62)
13	37	27 (73)

patients. Thus, this observation suggests a different pattern of in vivo TSPO binding between LAB and MAB + HAB patients. Although these data are consistent with the results of brain studies, the literature also reports that ^{18}F-DPA-714 PET/CT can visualize arthritic joints independently of polymorphism status, suggesting a lesser impact of TSPO polymorphism status on arthritis targeting.[18] The other patient, with low tumor contrast, had a TSPO MAB polymorphism but the smallest tumor in the sample, measuring only 9 mm on histological analysis, which may partly explain the lack of significant lesion uptake due to the partial volume effects.

TSPO expression in BC cell lines has been extensively studied by Zheng et al using DPA-714 autoradiography. The authors did not report any heterogeneity in radioactivity distribution between the tumor itself and its microenvironment but noted a strong correlation with F4/80-positive macrophage cell expression, an IHC biomarker of an inflammatory microenvironment.[20] More recently, Tuominen et al[14] reported the same pattern of DPA-714 binding in a head and neck cancer xenograft autoradiography study. In both studies, DPA-714 radiolabeled with fluorine-18 was used for autoradiographic purposes. Of interest, in the present study, we used tritiated DPA-714, which has a lower beta energy and is therefore more suitable for autoradiographic experiments, with improved resolution and quantification of the signal. This overexpression of TSPO in both the neoplastic and stromal compartments of the tumor could therefore contribute significantly to the sensitivity of the PET signal.

The ^{18}F-DPA-714 TACs obtained for TNBC patients showed 3 different kinetic patterns compared with the kinetics observed for the contralateral pectoral muscle. Ribeiro et al[27] has already shown, in a completely different pathology (cerebral stroke), that the kinetics of ^{18}F-DPA-714 differ between injured and normal tissues. Despite the small size of our patient group, some consistency with the patient TSPO polymorphism was found that no LAB patient presented a TAC as an “above muscular” curve. On the other hand, the HAB-MAB and MAB-LAB split observed with the TSPO polymorphism seemed less distinct when comparing ^{18}F-DPA-714 tumor uptake kinetics as some MAB patients’ TACs were similar to HAB patient TACs as “above muscular” uptake. These data are consistent with the results of a recent study showing a significant difference in the biodistribution of ^{18}F-DPA-714 depending on the TSPO polymorphism, which was highly and significantly decreased in LAB compared with HAB and MAB patients.[28] In addition, it takes a relatively long time to obtain the patient TSPO polymorphism status, which could be considered for patients with a neurodegenerative disease, but is incompatible with TNBC management, both at diagnosis and metastatic stage screening. Thus, in oncological studies, using second-generation TSPO tracers in PET/CT, and if it is not possible to obtain the TSPO polymorphism status before inclusion, it would be wise to exclude LAB patients from analysis a posteriori.

TME is playing an increasingly important role in oncology, particularly in understanding the mechanisms of tumor progression and resistance to treatment.[29] It has been observed that a high density of M2-type macrophages in breast tumors is strongly associated with rapid proliferation and poor differentiation.[30] Moreover, TNBC cells have been shown to secrete factors that promote macrophage polarization toward an M2 status.[31] However, we did not observe any significant correlation between ^{18}F-DPA-714 PET/CT uptake and macrophage tumor infiltration, whether in terms of numbers of pan-macrophages visualized using macrophage marker CD68 (IHC) or the percentage of M2-polarized macrophages using the marker CD163 (IHC), when compared with the semiquantitative parameters of ^{18}F-DPA-714 (SUV_{max}, SUV_{mean}, and %ID/g). Several factors could explain these results apart from the small size of our patient group, which limits the statistical power. Indeed, it is important to

TABLE 3. TSPO Polymorphism and Comparative Evaluation of TSPO Radioligands ^{3}H-DPA-714 and ^{3}H-PK-11195 by In Vitro Autoradiographic Binding on TNBC Tumor Slices

#	TSPO Polymorphism	(1) ^{3}H-PK-11195 SB (cpm/mm^2)	(2) ^{3}H-DPA-714 SB (cpm/mm^2)	Ratio (2)/(1)
1	High affinity binder (HAB)	–	–	–
2	Intermediate affinity binder (MAB)	6.16	2.86	0.46
3	High affinity binder (HAB)	–	–	–
4	Intermediate affinity binder (MAB)	5.20	3.31	0.64
5	Intermediate affinity binder (MAB)	9.10	6.97	0.77
6	High affinity binder (HAB)	7.85	5.83	0.74
7	Low affinity binder (LAB)	8.70	2.71	0.31
8	High affinity binder (HAB)	–	–	–
9	High affinity binder (HAB)	7.32	4.3	0.59
10	Low affinity binder (LAB)	10.60	3.69	0.35
11	Intermediate affinity binder (MAB)	8.75	4.72	0.54
12	Intermediate affinity binder (MAB)	15.04	6.79	0.45
13	Intermediate affinity binder (MAB)	13.01	9.61	0.74

note that TSPO is not specifically overexpressed in myeloid cells and macrophages. It has recently been observed that TSPO is also overexpressed, albeit to a lesser extent, in various immune cells such as neutrophils and T and B lymphocytes.[16] Additionally, TSPO is overexpressed in some breast tumor cells, particularly those that do not express hormone receptors and have a high Ki67.[11,13,32–34] In our study, we selected TNBC patients, and the pathological data from our patients showed a high median Ki67 of 60% (range, 13%–90%). Thus, TSPO is not specifically overexpressed in macrophages, which may explain the lack of correlation between ^{18}F-DPA-714 PET/CT binding intensity and macrophage infiltration in IHC. In addition, recent data suggest that increased TSPO expression in human central nervous system myeloid cells is related to the density of inflammatory cells rather than inflammatory activity, which must now be taken into account in the clinical interpretation of the TSPO PET signal, and potentially applies to extraneuronal myeloid cells.[35]

The overexpression of TSPO by BC cells is associated with tumor aggressiveness,[13] as well as the degree of TAMs infiltration with tumor progression and resistance to treatment.[6] This suggests that ^{18}F-DPA-714 uptake may be a potential prognostic indicator of tumor aggressiveness. ^{18}F-DPA-714 PET/CT may therefore become a valuable tool for identifying subgroups of patients at increased risk of tumor recurrence. Abnormal TSPO protein expression has been identified in numerous others tumor types, including astrocytoma, skin cancer, ovarian carcinoma, colorectal cancer, endometrial carcinoma, fibrosarcoma, and hepatocellular carcinoma, thus suggesting a role for TSPO in cell proliferation.[36] In glioblastoma, hepatocellular carcinoma, and melanomas as BC,[13,36–38] it is described that increased TSPO expression associates with poor prognosis. Recently, however, Tuominen et al[39] showed the opposite in head and neck squamous cell carcinoma where decreased TSPO expression was associated with poor prognosis. Previously, the same team showed changes in TSPO expression levels after external radiotherapy in head and neck squamous cell carcinoma, which does not seem to reflect inflammation. Most significantly, ^{18}F-DPA-714 detected these changes.[14] It is therefore likely that TSPO expression can vary between tumors, as can its prognostic value, which can also change under therapeutic pressure. ^{18}F-DPA-714 PET/CT can provide us with this knowledge, noninvasively, at the initial stage and during or after therapy.

TSPO has become a promising target for the development of new cancer treatments. Several approaches are being explored, including the use of anti-TSPO antibodies,[40] targeted drug delivery systems using TSPO ligands,[41,42] or photodynamic therapies targeting mitochondria using TSPO.[43,44] Thus, ^{18}F-DPA-714 PET/CT could be useful as a biomarker for the development of treatments directed against the TME and targeting TSPO. This "companion imaging" could allow better selection of patients likely to respond favorably to treatment and monitoring therapeutic response.

Our study has several limitations. The small patient group size probably contributed to the inability to reach statistically significant differences, especially regarding the correlation between macrophagic infiltration estimated by IHC and ^{18}F-DPA-714 PET/CT uptake, but it is a feasibility study. The decision to offer ^{18}F-DPA-714 PET/CT only to TNBC patients who were eligible for first surgery may have also affected our results, especially for tumors smaller than 1 cm. Nevertheless, the absence of prior therapeutic interventions enabled IHC and autoradiographic analyses to be performed on primary tumors without interference by treatment effects.

CONCLUSIONS

The results of this preliminary study confirm the potential of the TSPO radiotracer ^{18}F-DPA-714 to characterize TNBC tumors by PET/CT. We successfully demonstrated that ^{18}F-DPA-714 targets tumors with an excellent TBR. Dynamic PET/CT acquisition allowed us to differentiate ^{18}F-DPA-714 uptake patterns, potentially influenced by the TSPO polymorphism status, and which may represent a potential prognostic biomarker. If treatments targeting TSPO are developed, ^{18}F-DPA-714 PET/CT could be useful for therapeutic monitoring of TNBC as "companion imaging," which should be the subject of further studies.

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